

# Notes for Discussion: Low Effective Sample Size in the Card College-Proximity Application

Alexander Unger

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## 1 Motivation

In the Card college-proximity application, the outcome-weight diagnostics reveal extremely low effective sample sizes (ESS), especially for the treatment definition based on college completion,  $D = \mathbf{1}\{\text{education} \geq 16\}$ . This note summarizes my current interpretation of this pattern and lists discussion points for the meeting.

The main empirical observation is that the point estimates from DML-based and normalized kappa-based estimators are not necessarily wildly different, but the implied outcome weights are highly concentrated. This suggests that the estimates may be driven by a very small effective subset of observations, even though the nominal sample size is  $N = 3010$ .

## 2 Descriptive facts from the Card sample

The descriptive analysis gives the following basic sample structure:

- Total observations:  $N = 3010$ .
- Instrument  $Z = \text{nearc4}$ : 2053 observations, or 68.2%, live near a four-year college.
- Treatment  $D_1 = \text{somecol}$ : 1521 observations, or 50.5%, attended at least some college.
- Treatment  $D_2 = \text{educ16}$ : 817 observations, or 27.1%, completed college.
- Median years of education: 13.
- Mean years of education: 13.26.
- Black share: 23.4%.
- SMSA at survey: 71.3%.
- SMSA at age 16: 65.0%.

A first descriptive intuition is therefore that college completion is a much thinner treatment margin than college attendance. Roughly half of the sample attended some college, but only about one quarter completed college. This alone does not mechanically imply a low ESS, but it makes it plausible that the instrument-induced variation in college completion is much smaller than the instrument-induced variation in college attendance.

### 3 First-stage and complier-share intuition

For a binary instrument and a binary treatment, the raw first stage is

$$\Delta_D = \Pr(D = 1 \mid Z = 1) - \Pr(D = 1 \mid Z = 0). \quad (1)$$

Under the standard monotonicity assumption, this difference can be interpreted as the population share of compliers in the simple unconditional binary-IV setup. The intuition is:

$$\Pr(D = 1 \mid Z = 1) = \Pr(\text{always-taker}) + \Pr(\text{complier}), \quad (2)$$

$$\Pr(D = 1 \mid Z = 0) = \Pr(\text{always-taker}). \quad (3)$$

Thus,

$$\Pr(D = 1 \mid Z = 1) - \Pr(D = 1 \mid Z = 0) = \Pr(\text{complier}). \quad (4)$$

Observed treatment cells should be interpreted carefully:

| Observed cell  | Interpretation under monotonicity | Type directly identified?   |
|----------------|-----------------------------------|-----------------------------|
| $Z = 1, D = 1$ | always-taker or complier          | No                          |
| $Z = 1, D = 0$ | never-taker                       | Yes, in this realized state |
| $Z = 0, D = 1$ | always-taker                      | Yes, in this realized state |
| $Z = 0, D = 0$ | never-taker or complier           | No                          |

Therefore, individual compliers are not observed. What is identified is the excess treatment probability among those with  $Z = 1$ , relative to those with  $Z = 0$ .

### 4 Connection to the low ESS

The LATE can be written in its simplest Wald form as

$$\hat{\tau}_{\text{Wald}} = \frac{\hat{\mathbb{E}}[Y \mid Z = 1] - \hat{\mathbb{E}}[Y \mid Z = 0]}{\hat{\mathbb{E}}[D \mid Z = 1] - \hat{\mathbb{E}}[D \mid Z = 0]}. \quad (5)$$

The denominator is the first stage. If this denominator is small, the estimator divides by a small number. This mechanically amplifies noise and can generate extreme outcome weights. In the outcome-weight representation,

$$\hat{\tau} = \sum_{i=1}^N \omega_i Y_i, \quad (6)$$

a weak first stage implies that the weights  $\omega_i$  may become highly concentrated. The corresponding effective sample size is

$$ESS = \frac{1}{\sum_{i=1}^N \omega_i^2}. \quad (7)$$

A very low ESS therefore indicates that the estimate is effectively based on a very small number of high-leverage observations, even though the nominal sample size is much larger.

## 5 Current interpretation of the Card results

The most plausible interpretation is that `nearc4` is a more relevant instrument for college attendance than for college completion.

- For  $D_1 = \text{somecol}$ , living near a four-year college plausibly increases the probability of attending some college.
- For  $D_2 = \text{educ16}$ , the same instrument may have a much smaller impact. Proximity to a college may induce college entry, but it does not necessarily induce completion.
- Therefore, the complier margin for college completion is likely much smaller than the complier margin for college attendance.
- This smaller first stage makes the denominator of the LATE estimator smaller.
- A smaller denominator amplifies the implied outcome weights and can explain the collapse of ESS in the college-completion specification.

This interpretation is also consistent with the outcome-weight diagnostics: the very low ESS should not be interpreted as a small raw sample size problem, but as a weak-identification / limited-overlap problem in the instrument-induced treatment margin.

## 6 Main empirical finding to discuss

A concise formulation of the finding would be:

The Card application has a large nominal sample size, but the outcome-weight diagnostics suggest that the college-completion LATE is estimated from an extremely small effective sample. This seems to be driven by a weak first stage for the completion margin. While proximity to a four-year college may shift some individuals into attending college, it appears to shift far fewer individuals into actually completing college. Consequently, the estimated complier share for  $D = \text{educ16}$  is small, the LATE denominator is small, and the implied outcome weights become highly concentrated.

## 7 Discussion points for Prof. Knaus

1. Is the very low ESS primarily a weak-first-stage signal, an overlap problem, or a combination of both?
2. Should ESS be reported as a central diagnostic for LATE applications with outcome weights, especially when comparing DML and kappa estimators?
3. For Card's college-completion treatment, should the low ESS be interpreted as evidence that the estimand is empirically fragile, even if the point estimates are plausible?
4. Would it be useful to inspect the observations with the largest absolute outcome weights and report their covariate profiles?

5. Should the thesis distinguish between “attendance compliers” and “completion compliers” to explain why  $D_1 = \text{somecol}$  and  $D_2 = \text{educ16}$  behave so differently?
6. Can the outcome-weight diagnostics be framed as revealing a practical limitation of this empirical design that is not visible from point estimates alone?

## 8 Possible thesis wording

The following paragraph could be adapted for the empirical chapter:

The effective sample size diagnostics reveal a striking difference between the two treatment definitions in the Card application. Although the nominal sample contains 3010 observations, the implied outcome weights for the college-completion treatment are extremely concentrated. A natural explanation is the weakness of the first stage on the completion margin. Proximity to a four-year college appears more likely to affect whether an individual attends some college than whether he completes college. Under monotonicity, the raw first-stage difference can be interpreted as the instrument-induced complier share. A small first stage therefore implies that the LATE is identified from a narrow complier margin. Since the Wald-type estimand divides by this first-stage denominator, weak first-stage variation mechanically amplifies the outcome weights and reduces the effective sample size. The low ESS should therefore be interpreted not as a small nominal sample problem, but as evidence that the empirical LATE for college completion is highly leverage-driven.